[METRIC]
A-A-52515
August 14, 1995
SUPERSEDING
MIL-P-62426(AT)
15 August 1991
(see 5.5)

COMMERCIAL ITEM DESCRIPTION

POWDER, CHROMIUM CARBIDE-NICKEL CHROMIUM ALLOY (METRIC)

The General Services Administration has authorized the use of this commercial item description (CID) for all federal agencies.

1. SCOPE. This CID covers a two-component powder containing chromium carbide and nickel chromium alloy particles mechanically distributed in a uniform mixture (75 chromium carbide and 25 nickel chromium) used in plasma spray coating operations. This powder is primarily intended for use as an anti-fretting coating for use in environments up to 815 degrees Celsius (°C).

2. SALIENT CHARACTERISTICS

2.1 Materials. Unless otherwise specified herein, the materials used shall be in accordance with the manufacturer's materials specification. The use of recovered material made in compliance with regulatory requirements is acceptable providing all requirements of this CID are met (see 5.5).

2.1.1 Formulation. The two component mixture shall consist of the percentage by weight as shown in table I.

Beneficial comments, recommendations, additions, deletions clarifications, etc. and any other data which may improve this document should be sent by letter to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E, Warren, MI 48397-5000.

FSC 9630

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TABLE I. Component percentages by weight.

Component	Percentage (%)
A (Chromium carbide) B (Nickel chromium alloy)	74 - 76 Remainder

2.1.2 Particle size. The particle size distribution by weight shall be as shown in table II.

TABLE II. Particle size distribution by weight.

Sieve designation 1/		% by weight				Weight Passed through sieve	Weight Retained on sieve
		Chromium carbide		Nickel chromium alloy			
Micrometer	Mesh	Min	Max	Min	Max		
53	270	100	---	100	---	X	
45	325	---	1	---	1		X
20	---	60	90	45	75	X	
5	---		15	---	5	X	

1/ Sieve designation in accordance with ASTM E11.

2.1.3 Chemical composition. The chemical composition of the material shall conform to the percent by weight for each element as specified in table III.

TABLE III. Chemical composition.

Elements	% by weight			
	Chromium carbide		Nickel chromium alloy	
	Min	Max	Min	Max
Manganese	---	---	---	2.5
Silicon	---	0.1	---	1.5
Iron	---	0.7	---	1.0
Carbon	12.5	---	---	0.25
Chromium	85.5	---	18.0	22.5
Nickel	---	---	76.0	80.0

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2.1.4 Properties. The material shall meet the following properties when applied (plasma sprayed) to iron, and nickel-base alloys.

a. Bond strength. The bond strength of the plasma-sprayed material shall be not less than 31 megapascals (MPa).

b. Cup test. The amount of draw before spalling of the coating shall not be less than 5.1 millimeters (mm).

2.2 Unit containers. Unless otherwise specified (see 5.3), the unit containers shall be capable of holding the following quantities of powder: 0.5, 1.0, 2.5, 5.0 and 10.0 kilograms (kg) The containers shall be sealed moisture proof units and large enough to permit mixing the powder within before use. The containers shall be capable of protecting the powder against damage or contamination until it is ready to be used.

2.3 Identification markings. The identification markings shall be permanent and legible and shall include, as a minimum, the manufacturer's CAGE code, part number, and lot number and the part or identification number (PIN) (see 5.3).

2.3.1 Labeling. Unit containers and packages shall be conspicuously labeled "CAUTION: Containers contain powder metal. Use approved respirator."

2.3.2 Container. Each container shall be marked "Chromium carbide-nickel chromium alloy powder".

3. QUALITY ASSURANCE PROVISIONS

3.1 Responsibility for inspection. The contractor is responsible for the performance of all inspections (examinations and tests).

3.2 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description and that the product conforms to the producer's own drawings, specifications, workmanship standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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4. PRESERVATION, PACKAGING, PACKING, LABELING, AND MARKING

Preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order (see 5.2).

5. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

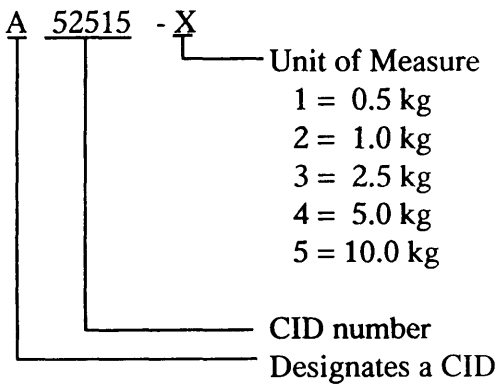
5.1 Addresses for obtaining copies of referenced documents.

5.1.1 Non-Government publications. Copies of ASTM E11 "Standard Specifications for Wire Cloth Sieves for Testing Purposes" are available from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.

5.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this CID.
b. Issue of Department of Defense Index of Specifications and Standards (DODISS) to be cited in the solicitation, and if required, the specific issue of individual document referenced.
c. Selection of applicable level and packaging requirements.
d. Unit container and PIN number.
e. Unit container size if other than specified.

5.3 Part or identification number (PIN). The PINs to be used for powder acquired to this CID, are created as follows:



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5.4 Cross reference data. Powder conforming to this CID is interchangeable/substitutable with powder conforming to MIL-P-62426(AT). Cross reference to former part number is as indicated in table IV.

TABLE IV. Cross reference.

Unit Container size	PIN	Former PIN No.
0.5 kg	A52515-1	M62426- 0.5 kg
1.0 kg	A52515-2	M62426- 1.0 kg
2.5 kg	A52515-3	M62426- 2.5 kg
5.0 kg	A52515-4	M62426- 5.0 kg
10.0 kg	A52515-5	M62426-10.0 kg

5.5 Supersession data. This CID supersedes Textron Lycoming (Allied Signal) internal specification M3965A, dated 26 April 1982.

5.6 Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable.

MILITARY INTERESTS:

Custodian:
Army - AT

CIVIL AGENCY COORDINATING ACTIVITY:
GSA-FSS

PREPARING ACTIVITY:
Army - AT

(Project 9630-0032)

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